NIHVA
NIHVA Technologies Pvt. Ltd.

SYSTEM & MAINTENANCE MANUAL
FOR
ELECTRO HYDRAULIC SCISSOR LIFT

CLIENT: DECON INDIA PVT LTD.

MFG.YEAR: - FEB 2019.

SERIAL NUMBER: - NSCL327

DESIGNED, DEVELOPED & MARKETED

BY,

NIHVA
NIHVA Technologies Pvt. Ltd.

G-BLOCK, 1ST FLOOR, THERMAX CHOWK
MIDC, ABOVE RENAULT SHOWROOM
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SCISSOR LIFT	
SIZE	2000 mm X 1500 mm
CAPACITY	6000 KG

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Method statement for pit mounted scissor lift:

1. Check all pit dimensions for accuracy.
2. Attach a temporary electrical line through the pit conduit to the lift. Check for correct motor rotation
3. Using slings, encircle the entire lift, not just the platform and lower the lift into the pit, centering it for 1” minimum clearance on all sides to the pit wall.
4. Raise the lift with the push button and remove the slings. Run the unit up and down several times to remove air from the hydraulic system.
5. Level and center the lift by shimming and grouting the entire base frame, not just the corners. Lag the unit in place using M16 x 150L wedge anchors in the holes provided. Caution! Continuous base frame support is essential for proper installation
6. With the lift fully elevated, disconnect the main power and complete the permanent electrical wiring.
7. Follow the instructions outlined in operating manual to complete the installation.

1. GENERAL INFORMATION

1. The hydraulic scissor lift load capacity rating and the serial number is mentioned on metal name plate.
2. Lift should not be overloaded beyond the rated load capacity may result in damage.
3. The hydraulic system consists of pump, motor, oil tank & valves.
4. The gear pump is set at operating pressure of 200 bar.
5. The operating principle is that the pump is operated to raise the table, and the pump is stopped when the table attains the desired elevation. A check valve and lowering valve between the pump and cylinder holds the table at elevation. For lowering to any desired level, the solenoid valve is energized to allow fluid to return from hydraulic cylinder to reservoir. A pressure compensated flow control valve is connected to the hydraulic cylinder to control the down speed at a predetermined rate under all load conditions.
6. A scissor lift should only be operated on a solid surface such as concrete, tile, or a compacted road base. When lifts are placed on uneven surfaces or earth, mulch or gravel that is not compacted, the lifts may tip and fall over, causing serious injury or death to the people on the platform.

1.1 MAIN SPECIFICATIONS

A) Maximum Load Lifting Capacity	6000 KG
B) Max Height of Platform	2800 MM
C) Actual Lift	2000 MM
D) Min Height of Platform	800 MM
E) Platform Length	2000 MM
F) Platform Width	1500 MM
G) Motor Capacity	5 HP
H) Oil Tank Capacity	80 LTR

2. ELECTRO HYDRAULIC SCISSOR LIFT- DO's & DONT's: -

DOs: -

- Check all the electrical connections.
- Check the hydraulic connections.
- Ensure NO Oil Leakage from joints.
- Ensure the oil level on level indicator before using the lift.
- Before loading & unloading ensure the lip rests on truck properly.
- Ensure that there isn't any obstacle within the region of moving parts.
- While doing maintenance work, ensure that the Emergency switch is pressed & the maintenance stoppers are in their position while working below the lift.
- During brake down condition, use the LOTO switch.

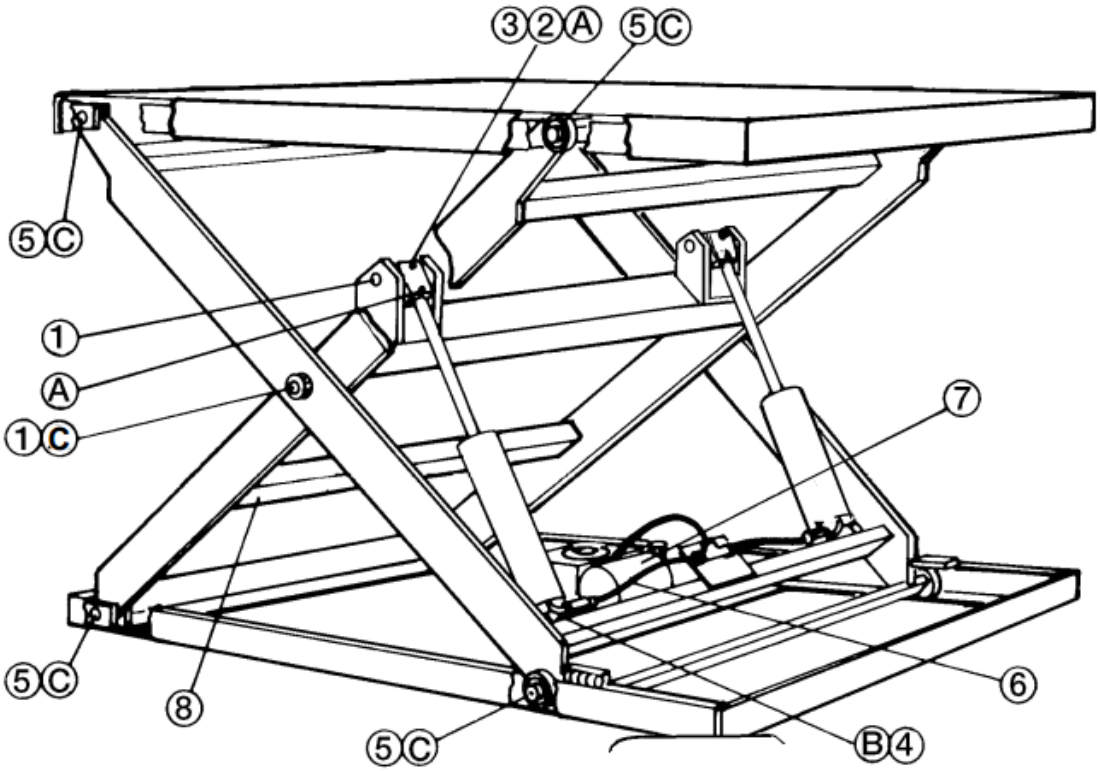
DON'Ts: -

- Lift should not be loaded beyond its rated load capacity.
- Don't lose the oil hoses without resting the lift on its maintenance stoppers.
- Do not operate damaged lift.
- Don't stand at the edges of lift platform while the lift is lowering.
- Don't change lowering valve setting without supervision or any assistance.
- In case of power failure, to resume the downward operation after Power ON, do not lower the lift. Instead, take the lift slightly upward & then resume to lower it.
- Don't put any manual stopper or anything in the roller track.

3. OPERATING INSTRUCTIONS

a) Check all electrical connection for short or phase short. Check hydraulic connections are tightened properly to ensure no oil leakage from joints.
b) Connect (3 phase 415V) supply to the panel through cable top.
c) Switch ON the RED LOTO SWITCH on panel & ensure Emergency push button is in unlock position.
d) Use Lift's PUSH BUTTONS for moving lift upward & downward.
e) Use Lip's PUSH BUTTONS for moving lip upward & downwards.
f) In case of emergency, press emergency STOP push button provided on push button station.
g) Before lifting Scissor, check motor rotates in clock wise direction.
h) For controlling the lowering speed, adjust the valve provided on manifold.
i) In case of failure & scissor lift platform is at top position, loosen the threads of manual override provided in the solenoid valve after reaching the platform at down position again retighten the same.

4. LUBRICATION & ROUTINE MAINTENANCE INSTRUCTIONS



- **Lubrication Points**

Lubrication points as mentioned below,

A. Points noted with ‘A’, apply grease lubrication.

B. Points noted with ‘B’, apply oil lubrication.

C. Points noted with ‘C’, no lubrication required (Pre-Lubricated &sealed)

Ensure all Moving parts like Wheel, Flap Joint, Rollers etc. are greased.

- **Inspection Check Points**

a) Visually inspect all jack pins, pivot pins to ensure that the set screw has not loosened, allowing pins to move.

b) Confirm Hex Nut Bolt, Allen Bolt -Pivot Pin Locking are Fixed firmly.

c) Confirm Grub screw, Cup Washer on Scissor Arm Joint Bolt are fixed & are tightened firmly.

d) Confirm Tightening of Rollers & its free movement.

e) Confirm all Bolts at Scissor Arm are tightened.

f) Visually inspect all lift arm rolling bearings to ensure that bearing bolts are tight.

g) Visually inspect all hydraulic hoses and fittings to ensure there are no cut on hoses or leaking fittings.

h) Visually inspect all electrical lines to ensure there are no cut wires or loose cords.

i) Visually inspect the entire lift structure.

j) These recommendations are applicable to the equipment considering daily usage of approx. 15-to 20 times.

If usage is more frequent, the same time intervals for maintenance should be adjusted accordingly.

- **WEEKLY**

a) Do greasing / oiling of all moving joints.

- **MONTHLY**

a) Check hydraulic pipeline for leak proof working.

b) Check oil level in hydraulic tank.

c) Check cylinder seals & ‘O` rings for leak proof working.

- **YEARLY**

a) Change hydraulic oil.

b) Check main shaft seal & shaft ‘O` ring for leak proof working.

Hydraulic oil to be used – Enclo – 68 OR Servo – 68.

5. GUIDE LINES IN CASE OF OIL LEAKAGE

a) Leakage at Piping –

- Tighten Joints.
- Replace Dowty Seal at Mounting

b) Leakage in Cylinder-

- Replace Seals.

c) Replace Oil after Every Year-

- Drain the Oil from the System.
- Clean filter, Tank elements, and strainers before refilling.

6. TROUBLE SHOOTING GUIDELINES

i. If lift will not raise:

a. Check electric circuit on new installations of 3-phase motors, if lift does not start raising in approximately 10 seconds, reverse two power supply leads to reverse pump rotation. (if motor direction is reverse)
b. Check oil level. With lift in full up position, oil level should be Approximately 1 1⁄4" above tank bottom.

ii. ***Lift raises slower than specified rate or will raise only partial load:***

Note: Allowance must be maintained for slight speed variations (due to temperature which changes the viscosity of the hydraulic oil varies)
a. Check line voltage under load condition. Low voltage affects capacity of equipment.
b. If foam is visible in tank oil, check for a loose connection in suction line between pump and tank. Tighten connections.

iii. Lift slowly settles with lowering valve not energized:

a. Flush lowering valve by cycling lift up and down. This should be done with no load.
b. If lift continues to settle, replace down valve cartridge.
c. If lift continues to settle, replace pump.

iv. If Lift will not lower:

a. Look for mechanical restrictions.
b. Test valve coil for operation. Check voltage at coil.
c. If voltage is ok, replace coil.

v. If lift will not raise to full vertical travel:

a. Check for low oil level if necessary, add oil.

vi. If lift does not lower smoothly:

a. Run the lift up and down under load condition for some time to purge air.

7. GENERAL SAFETY INSTRUCTIONS

Use trained personnel for loading & unloading of equipment. The equipment must be properly handled when transferring from the unloading area to final site to prevent damage.

DO NOT operate equipment unless it is properly anchored. It may cause serious injury or death.

a) Install all safety devices, guards and guarding prior to equipment start-up.
b) Only Trained & Authorized person must use the lift.
c) Always use mechanical stopper if you are going to work in elevated condition.
d) Keep working area free from oil to Avoid Slipping/Fire hazards.
e) Do Not Exceed the Lift Capacity than rated.
f) Before cleaning or servicing disconnect power supply
g) As the lift moves up and down, “pinch points” are formed. Keep hands, feet, hair, loose clothing, etc. away from these pinch points. Any of these may cause serious personal injury or death.
h) Ensure that the hydraulic, pneumatic and electrical lines will not be pinched by the lift as it rises or lowers. If a line gets pinched, the lift may not work properly. A hose may break, the lift table may drop suddenly,

and someone may get injured. If an electrical line is broken, may cause personal injury or death due to electrical shock.

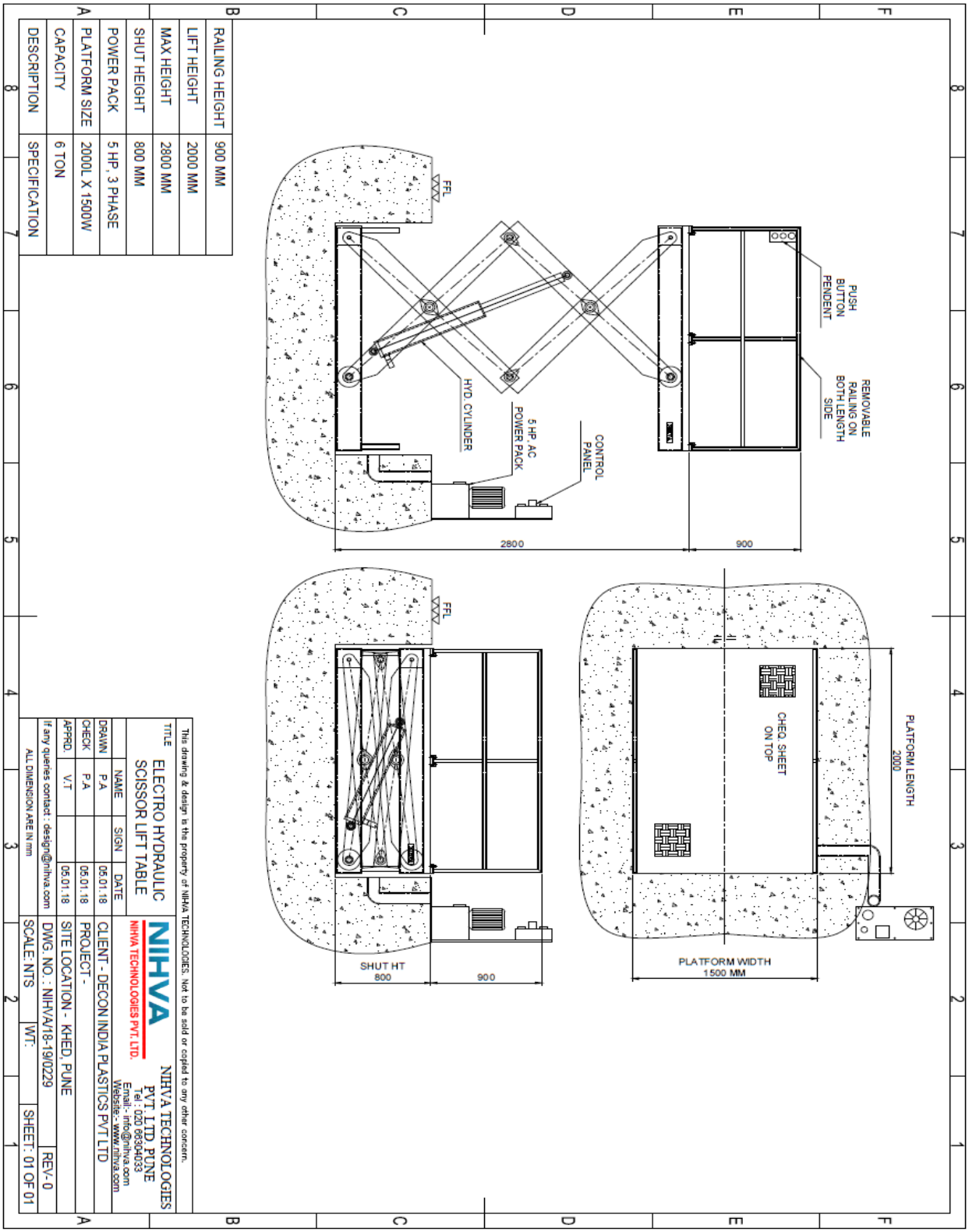
i) Read the motor nameplate and record the motor full load current. Consult local electrical codes for power supply fusing requirements. To avoid fire hazards, follow these fusing requirements.



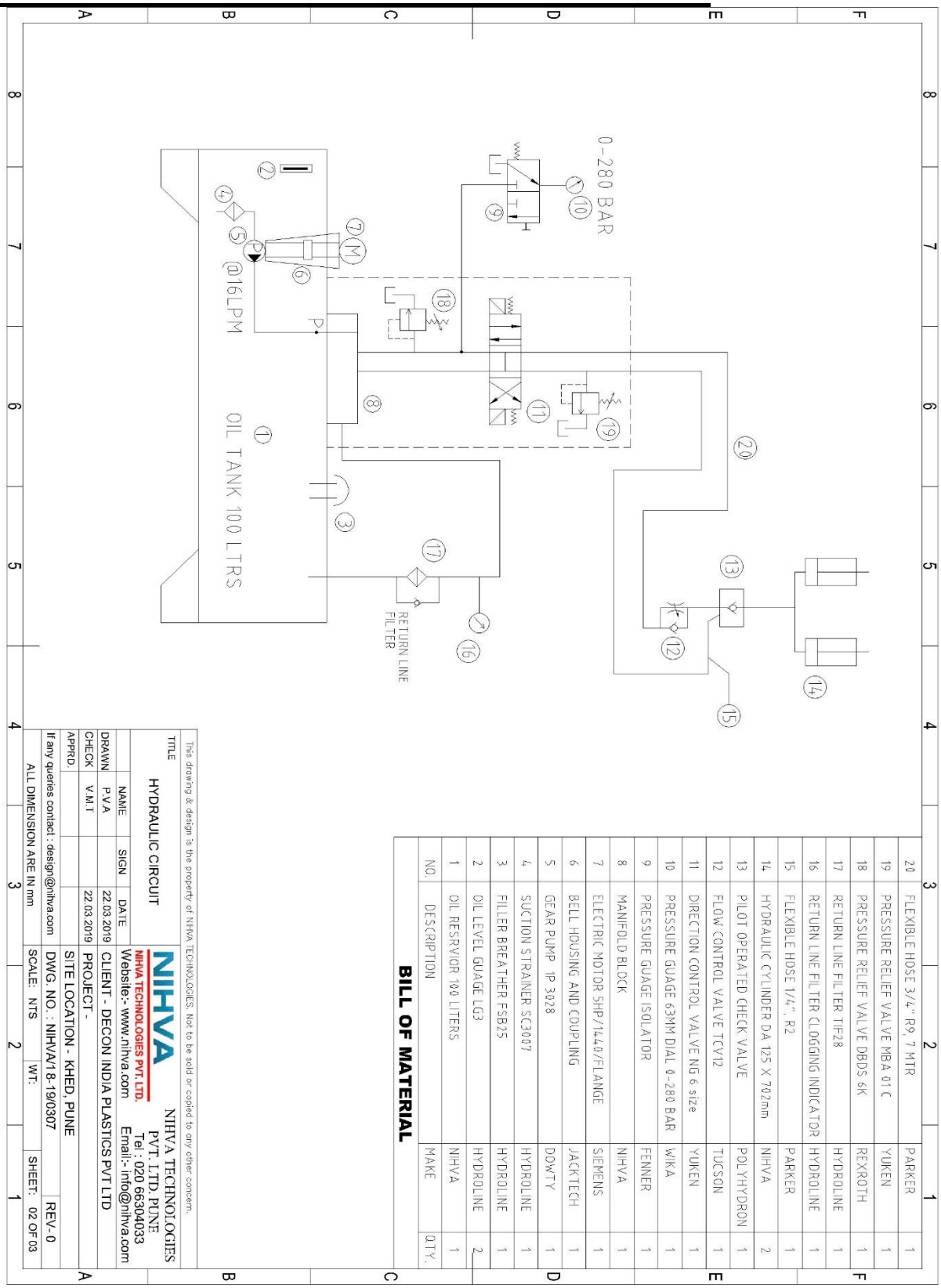




8. GA DRAWING:-


RAILING HEIGHT 900 MM
LIFT HEIGHT 2000 MM
MAX HEIGHT 2800 MM
SHUT HEIGHT 800 MM
POWER PACK 5 HP, 3 PHASE
PLATFORM SIZE 2000L X 1500W
CAPACITY 6 TON
DESCRIPTION SPECIFICATION
FFL
PUSH BUTTON PENDENT
REMOVABLE RAILING ON BOTH LENGTH SIDE
HYD. CYLINDER
5 HP, AC POWER PACK
CONTROL PANEL
2800
900
CHEQ. SHEET ON TOP
PLATFORM LENGTH 2000
PLATFORM WIDTH 1500 MM
SHUT HT 800
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TITLE
ELECTRO HYDRAULIC SCISSOR LIFT TABLE
NIHVA TECHNOLOGIES PVT. LTD.
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Tel.: 020 66304033
Email:- info@nihva.com
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NAME SIGN DATE
DRAWN P.A 05.01.18
CHECK P.A 05.01.18
APPRD V.T 05.01.18
CLIENT - DECON INDIA PLASTICS PVT LTD
PROJECT -
SITE LOCATION - KHED, PUNE
If any queries contact : design@nihva.com
DWG. NO. : NIHVA/18-19/0229
REV- 0
ALL DIMENSION ARE IN mm
SCALE: NTS
WT:
SHEET: 01 OF 01

9. HYDRAULIC POWER PACK & CIRCUIT: -



BILL OF MATERIAL

NO.	DESCRIPTION	MAKE	QTY.
20	FLEXIBLE HOSE 3/4" R9, 7 MTR	PARKER	1
19	PRESSURE RELIEF VALVE MBA 01 C	YUKEN	1
18	PRESSURE RELIEF VALVE DBDS 6K	REXROTH	1
17	RETURN LINE FILTER TIF28	HYDROLINE	1
16	RETURN LINE FILTER CLOGGING INDICATOR	HYDROLINE	1
15	FLEXIBLE HOSE 1/4", R2	PARKER	1
14	HYDRAULIC CYLINDER DA 125 X 70Zmm	NIHVA	2
13	PILOT OPERATED CHECK VALVE	POLYHYDRON	1
12	FLOW CONTROL VALVE TCV12	TUCSON	1
11	DIRECTION CONTROL VALVE NG 6 size	YUKEN	1
10	PRESSURE GUAGE 63MM DIAL 0-280 BAR	WIKA	1
9	PRESSURE GUAGE ISOLATOR	FENNER	1
8	MANIFOLD BLOCK	NIHVA	1
7	ELECTRIC MOTOR 5HP/1440/FLANGE	SIEMENS	1
6	BELL HOUSING AND COUPLING	JACKTECH	1
5	GEAR PUMP 1P 3028	DOWTY	1
4	SUCTION STRAINER SC3007	HYDROLINE	1
3	FILLER BREATHER FSB25	HYDROLINE	1
2	OIL LEVEL GUAGE LG3	HYDROLINE	2
1	OIL RESRVOIR 100 LITERS	NIHVA	1

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TITLE: HYDRAULIC CIRCUIT

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	NAME	SIGN	DATE
DRAWN	P.V.A		22.03.2019
CHECK	V.M.T		22.03.2019
APPRD.			

If any queries contact : design@nihva.com

CLIENT - DECON INDIA PLASTICS PVT LTD

PROJECT -

SITE LOCATION - KHED, PUNE

DWG. NO. : NIHVA/18-19/0307

REV- 0

ALL DIMENSION ARE IN mm | SCALE: NTS | WT: | SHEET: 02 OF 03

10. ELECTRICAL CIRCUIT:-

